



REVIEW - 1

Smart Attendance System Using Facial Recognition

Guide: KHAJA SHAREEF

SECTION 12 • DBS & DB

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Project Overview

What the system does and why it is needed

PROJECT IDEA

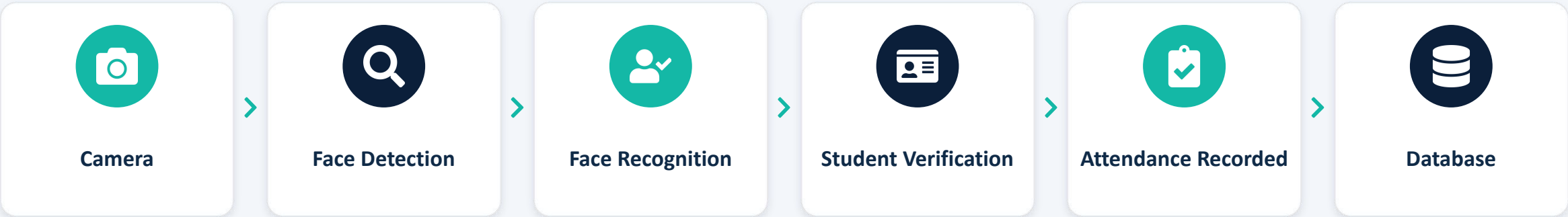
An automated attendance management system that uses a camera to identify registered students through facial recognition and record attendance in real time.

It reduces manual work, improves record accuracy and makes monitoring easier for administrators.

REVIEW 1 FOCUS

- Database structure and storage
- System architecture and modules
- Attendance workflow
- Admin monitoring and reports
- Implementation and testing plan

CORE FLOW



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Problem Statement & Proposed Solution

Why manual attendance falls short, and how we address it

CURRENT CHALLENGES

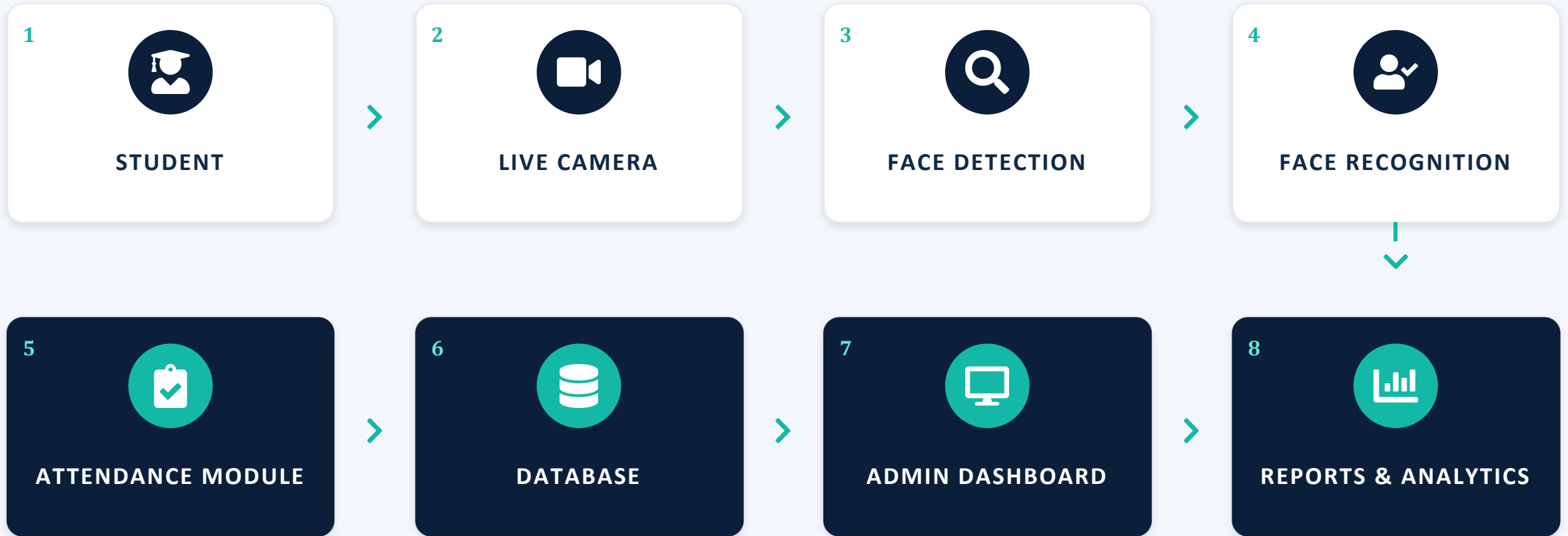
- ✗ Manual attendance takes time and effort
- ✗ Repetitive data entry can cause errors
- ✗ Large classes are difficult to manage
- ✗ Paper / spreadsheet records are hard to monitor
- ✗ Manual report preparation adds extra work
- ✗ Proxy attendance is difficult to control

PROPOSED APPROACH

- ✓ Detect faces through a live camera
- ✓ Match registered students with facial data
- ✓ Mark attendance automatically after verification
- ✓ Store attendance digitally with date and time
- ✓ Provide an admin interface for monitoring
- ✓ Generate attendance reports and analytics

3 System Architecture

High-level components and data flow



The architecture connects real-time recognition with digital attendance storage and administrative monitoring.

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Database Design – Proposed Structure

Logical structure for student, facial and attendance information

STUDENTS

- **student_id**
- name
- roll_no
- class/section
- contact/status

FACE_DATA

- **face_id**
- student_id
- face_identifier
- image/reference

ATTENDANCE

- **attendance_id**
- student_id
- date
- time
- status

ADMIN

- **admin_id**
- name
- login/authentication



Key relationship: one registered student can have facial data and multiple attendance records.

Proposed logical structure for Review 1; final fields can be refined during implementation.

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Technology Stack

Technology listed in the current Review 1 material



FRONTEND

HTML, CSS, JavaScript

User-friendly attendance interface



BACKEND

Python + Flask

Application logic, requests and component communication



AI / VISION

OpenCV + Face Recognition

Real-time face detection, recognition and identification



DATABASE

MongoDB

Student profiles, face identifiers and attendance records



CONNECTIVITY

PyMongo

Connects Flask backend with MongoDB



HARDWARE

Webcam / CCTV

Captures faces in real time

Development tool: Visual Studio Code — coding, testing and project development

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Review 1 Implementation & Testing

Work to demonstrate during the review

IMPLEMENTATION PROGRESS / TARGETS

- ✓ Finalize database structure and collections
- ✓ Connect application layer with database
- ✓ Implement student registration flow
- ✓ Integrate face detection and recognition
- ✓ Implement automatic attendance marking
- ✓ Build admin monitoring interface
- ✓ Add report / analytics views

TESTING PLAN

- 🧪 Test valid registered face recognition
- 🧪 Test unknown / unregistered face handling
- 🧪 Verify correct student is linked to attendance
- 🧪 Check date and time recording
- 🧪 Test duplicate attendance handling
- 🧪 Verify database records after attendance
- 🧪 Test admin viewing and report generation

7 Expected Outcome & Future Enhancements

EXPECTED OUTCOME

A working attendance system that identifies registered students, records attendance digitally, maintains database records, and gives administrators monitoring and reporting features.

EXPECTED BENEFITS

Less manual effort

Faster attendance process

Better record management

Reduced attendance errors

Easier monitoring

FUTURE ENHANCEMENTS

- Improve recognition under different lighting
- Add richer analytics and visual reports
- Add attendance export options
- Strengthen access control and data protection
- Support larger student datasets
- Improve usability of the admin dashboard

Thank You

Questions & Feedback Welcome

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